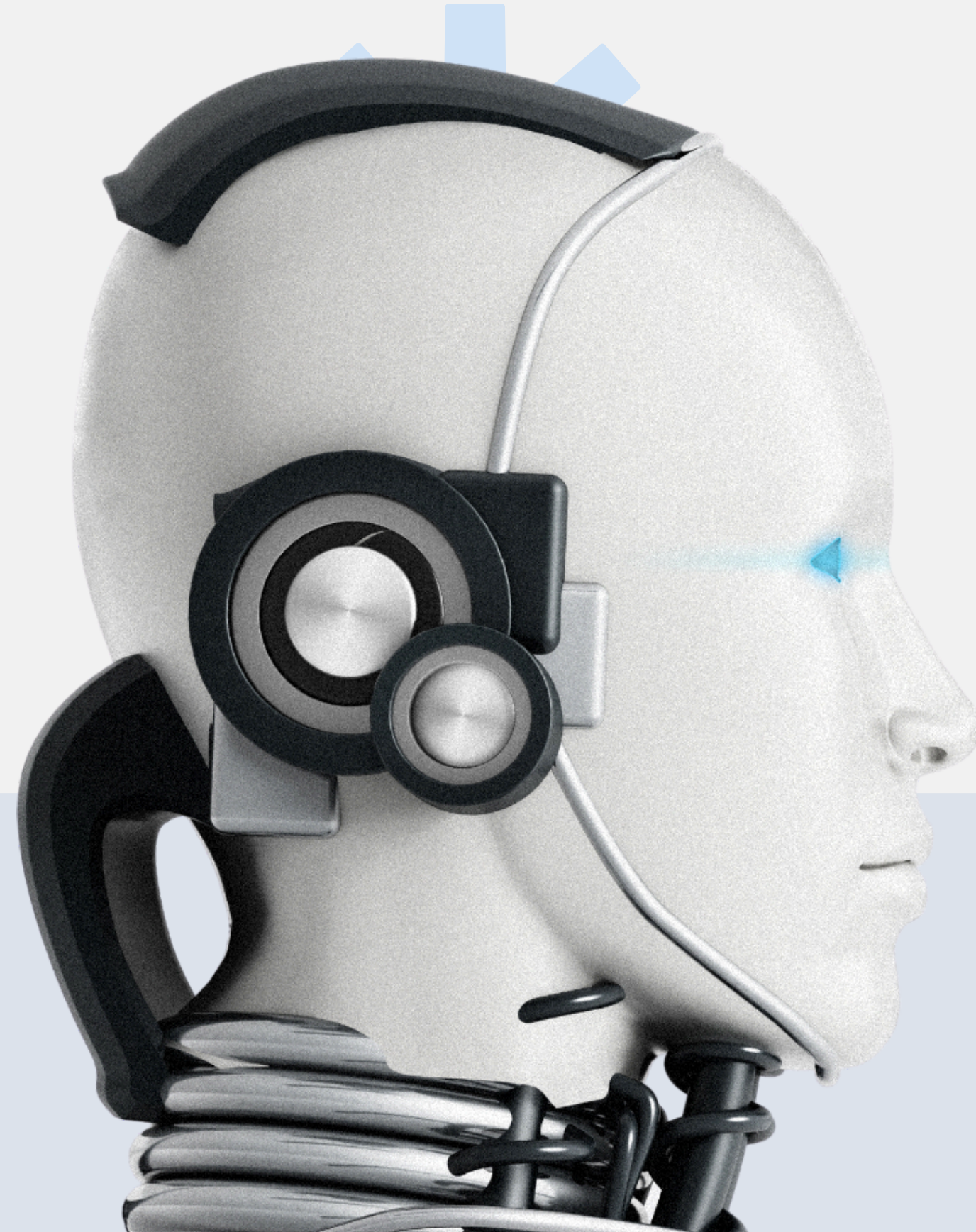


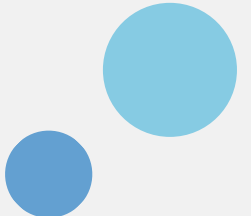
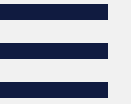
Machine learning

TRAFFIC FLOW PREDICTION

Done by:

ANUSHA-MS235108
AFIFA-MS235102



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ABSTRACTION

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- Traffic congestion is one of the biggest issues in cities, leading to delays, fuel loss, stress, and more pollution.
- To solve this, predicting traffic flow in advance can help in better planning and management of roads.
- Traffic flow prediction uses data from sensors, GPS, and cameras placed on roads to understand vehicle movement.
- Machine Learning (ML) provides powerful methods to analyze this data and forecast traffic conditions.
- The project focuses on applying different ML models to predict short-term traffic flow.
- Models like Linear Regression, Random Forest, and Support Vector Regression are tested for performance.
- Deep learning models, especially Long Short-Term Memory (LSTM) networks, are also explored since they handle time-based data well.
- The process involves cleaning the data, selecting useful features, and training the models.
- Evaluation is done using accuracy measures like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE)..



CHOSEN ARCHITECTURE & JUSTIFICATION

ARCHITECTURE SELECTED

- CNN → extract spatial/local patterns
- BiLSTM → capture temporal dependencies
- Residual Correction (ARIMA-inspired) → handle linear trends/seasonality
- Dense Layer → final traffic count prediction

JUSTIFICATION

- Combines spatial, temporal, and linear pattern modeling.
- Robust for both short & long-term forecasting.
- Improves accuracy vs. standalone models.
- Scalable and easy to implement in TensorFlow/PyTorch.

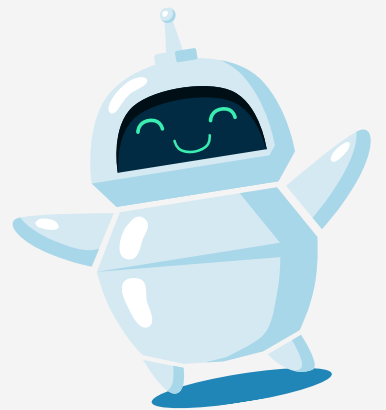


LITERATURE SURVEYS:



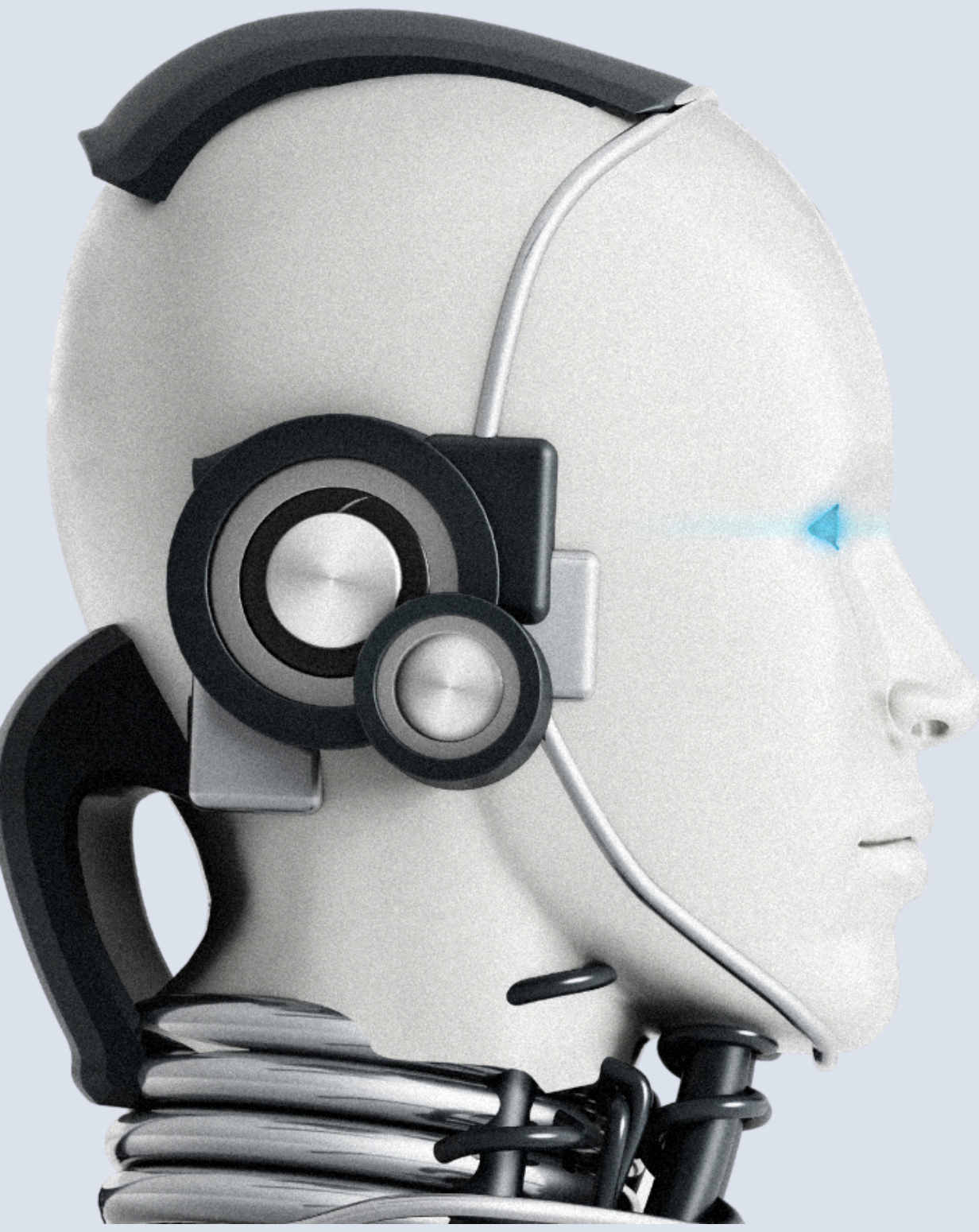
1) TRAFFIC PREDICTION USING LSTM, RF AND XGBOOST (DAML 2024)

- Authors & venue: Ka Nam Lam. Proceedings of the 2nd International Conference on Data Analysis and Machine Learning (DAML 2024).
- What they did (data & task): Predicted vehicle counts per hour at four city junctions using a Kaggle dataset (48,120 rows; features include DateTime, Junction ID, and Vehicles). Target was the next-hour vehicle count.
- Models compared: Random Forest, XGBoost, and LSTM.



Key findings:

- XGBoost outperformed RF and LSTM on MAE/RMSE/R² and was the most efficient in this setup.
- LSTM underperformed without richer features/tuning; authors note adding exogenous variables (weather/events) would likely help.



Architecture (*LSTM baseline*):

Add Time-series framing on hourly counts + calendar features (year/month/day/hour/weekday).

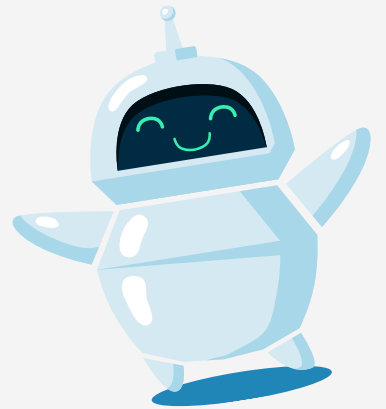
A single/two-layer LSTM to ingest the sequence window, followed by a Dense output layer to regress the next-hour count; standard training with tuned sequence length and basic regularization. (The paper describes the LSTM at a high level; exact hidden sizes aren't fixed in text.)

Source: Open-access PDF.



2) LONG-TERM TRAFFIC FLOW FORECASTING USING A HYBRID CNN-BILSTM MODEL (EAAI 2023)

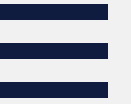
- Authors & venue: Manuel Méndez, Mercedes G. Merayo, Manuel Núñez. Engineering Applications of Artificial Intelligence, 121 (2023), Article 106041.
- What they did (data & task): Focused on traffic flow/vehicle-count forecasting over longer horizons on urban routes, where accuracy typically degrades versus short-term tasks.



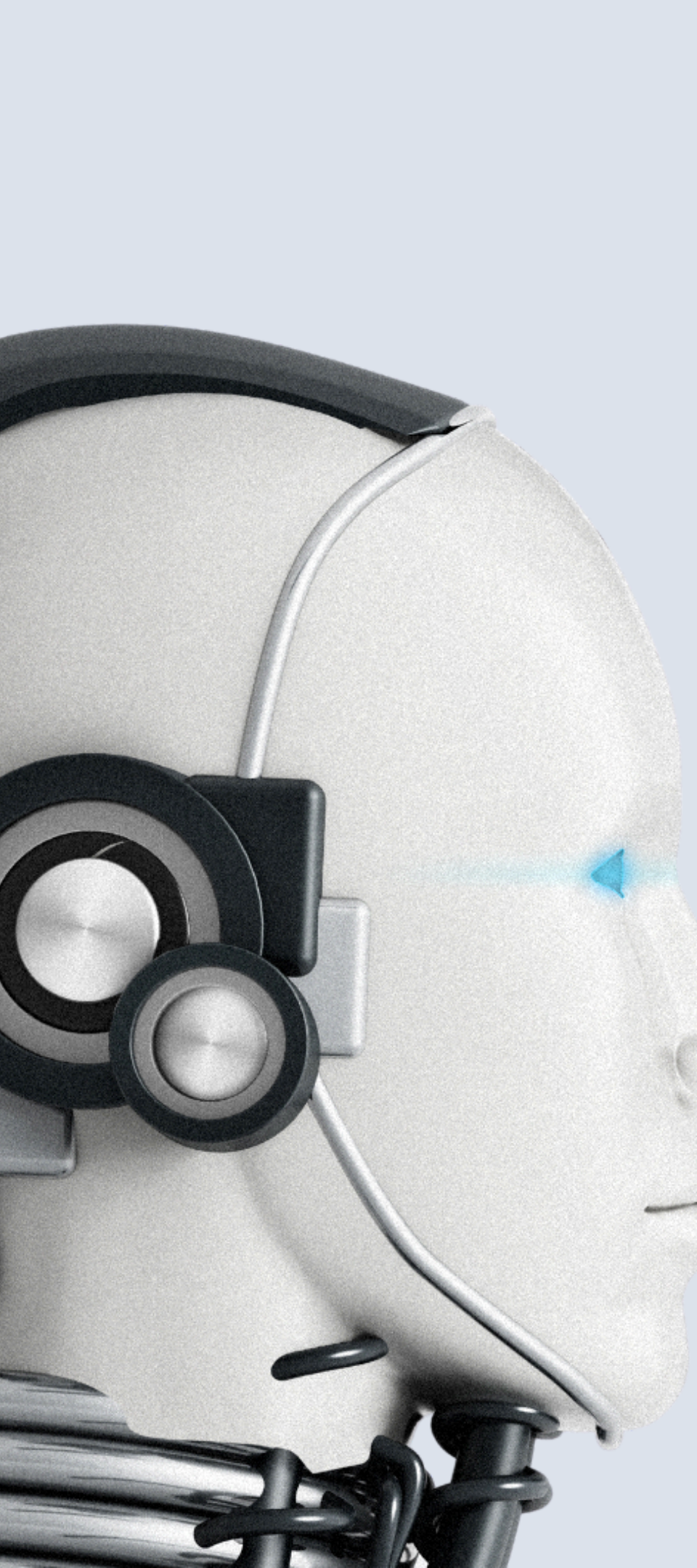
Key findings:

- A hybrid CNN→BiLSTM pipeline improved long-horizon forecasting versus strong baselines by capturing spatial patterns first and temporal dependencies next.
- Demonstrated robust gains for urban (noisier) settings.

Architecture (CNN-BiLSTM):



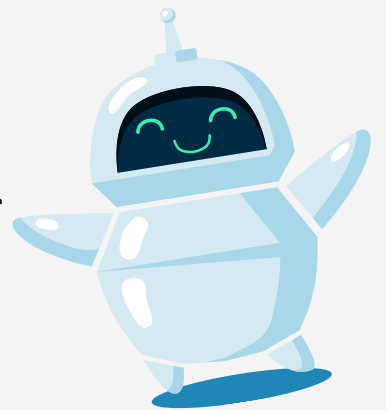
- Input: multivariate sequences of traffic counts (and related features) arranged as temporal windows.
- CNN block: 1D convolutions (with pooling) to extract local/spatial/multi-feature patterns from each window.
- BiLSTM block: bidirectional LSTM layers to capture past and future context within the window.
- Dense regressor: fully connected head maps BiLSTM embeddings to future traffic counts. (Paper describes the stacked hybrid design; exact layer widths are dataset-tuned.)
- Source: Elsevier/ScienceDirect article record (with article number) and abstract mirrors.



3) A COMBINED METHOD FOR SHORT-TERM TRAFFIC FLOW PREDICTION BASED ON ARIMA + LSTM (ALEXANDRIA ENGINEERING JOURNAL)



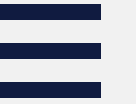
- Authors & venue: Saiqun Lu, Qiyan Zhang, Guangsen Chen, Dewen Seng. Alexandria Engineering Journal (article accepted 2020; AEJ volume lists around 2020–2021).
- What they did (data & task): Short-term vehicle-flow prediction; proposed a hybrid model that fuses classic time-series modeling with deep learning to handle both linear and nonlinear components.



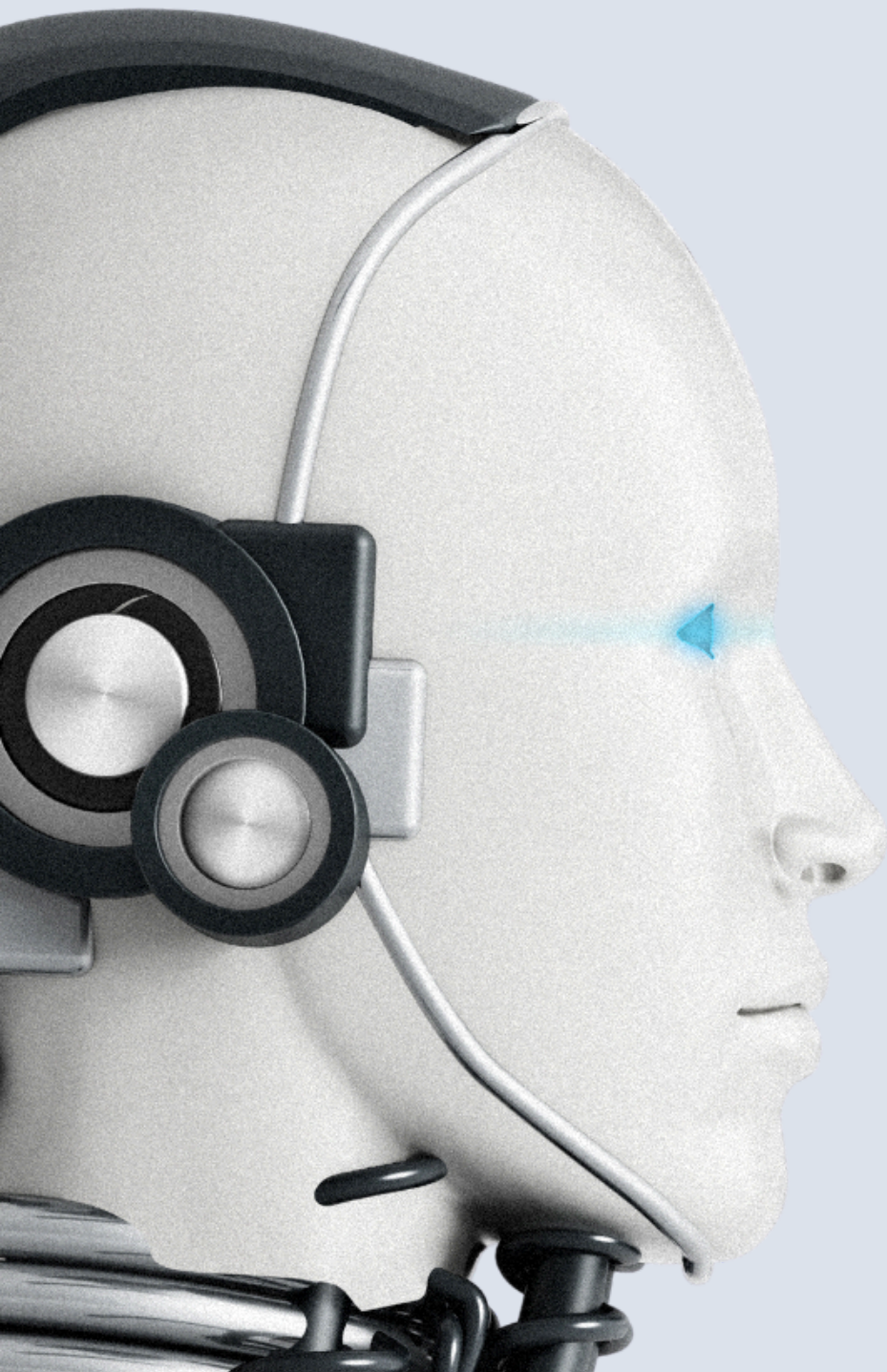
Key findings:

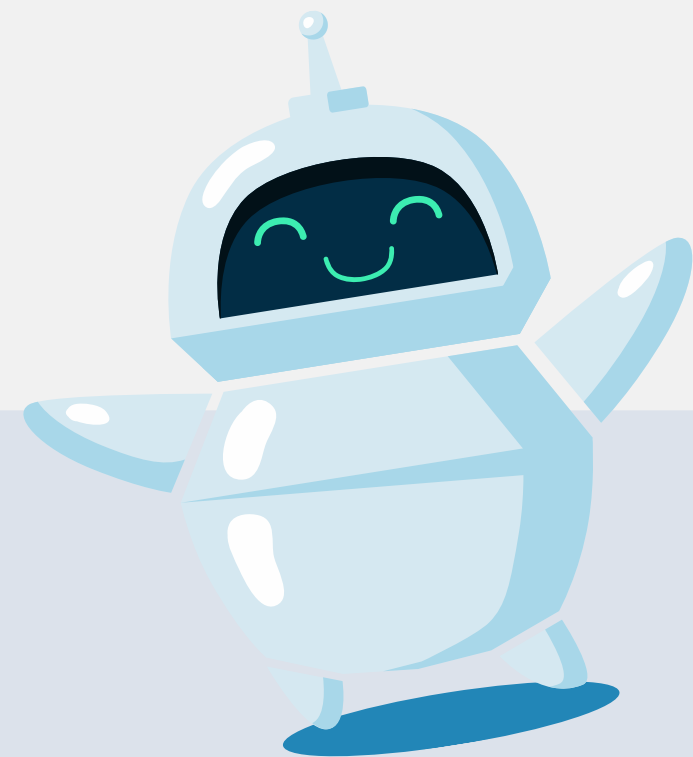
- The hybrid ARIMA + LSTM consistently outperformed standalone ARIMA or standalone LSTM across typical benchmarks of short-term flow.
- The decomposition (linear vs. residual nonlinear) streamlined learning and reduced error.

Architecture (ARIMA + LSTM):



- Stage 1 (ARIMA): Fit ARIMA to the vehicle-count series to model linear/stationary structure; produce a one-step-ahead forecast and residual sequence.
- Stage 2 (LSTM on residuals): Train an LSTM on residual sequences (and optionally exogenous features) to learn nonlinear temporal patterns.
- Fusion: Final prediction = ARIMA forecast + LSTM residual forecasts
- Source: Article record and accepted-manuscript PDF listing (ResearchGate/Elsevier entry).





THANK YOU

